TEST PROJECT

Skills Challenge 2026

AUTONOMOUS MOBILE ROBOTICS

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EmiratesSkills مهارات الإمارات

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INTRODUCTION

Technologies of this module: Robot built implementing the Studica kit.
Time to complete: 3 hours.

The primary focus of the Skills Challenge Test Project is to outline the requirements for teams to design, build, program, and operate a Mobile Robot to interpret the environment and tasks introduced at the competition. It is expected that competitors will use the information provided in this document to manage both their overall robot mobility and their object-handling experiences with the OMS.

DESCRIPTION OF PROJECT AND TASKS

Teams of two competitors are required to develop an autonomous mobile robot that efficiently operates in a known environment that represents real-world scenarios or tasks and can manipulate specific objects in both autonomous and tele-operated modes.

The test project is broken down into three assessment scenarios:

1. Basic robot functionality (simple tasks).
2. Robot performance in teleoperated control mode.
3. Robot performance in autonomous control mode.

Basic robot functionality:

1. Robots should be able to execute simple drive commands such as drive forward, drive backward, rotate, and move the gripper up/down and open/close.
2. Robots should be able to execute a group of commands such as drive forward 1 meter and then rotate 90 degrees left.
3. Robots should be able to execute a group of commands, such as drive forward, move up the gripper, close it, and then drive the robot backward.
4. Robots should be able to display the object's color in the camera FOV in the shuffleboard.

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Challenge: The specific tasks that the robot will have to perform will be given to the competitors during the morning briefing.

Robot performance in teleoperated and autonomous control mode:

1. The robots should be able to interact with their environment by moving to different areas of interest (storage areas, delivery areas, and a hazardous materials area). Additionally, robots must interact with objects of different colors (representing a type of material), which must be transported to the area corresponding to their color.
2. In teleoperated mode, competitors will interact with the robot through a direct line of sight, allowing them to identify the object's color without the implementation of a camera. The operator of the remote control can walk around the court so that they can see the entire site to assess the robot's performance.
3. In autonomous mode, the camera will play an important role in identifying the color of objects for their subsequent classification. The camera should be able to recognize a surprise color for an object representing a hazardous material.

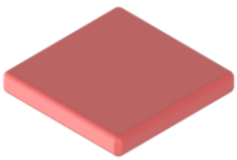
Challenge: Specific delivery instructions and the color of the hazardous object will be given to the competitors during the morning briefing.

INSTRUCTIONS TO THE COMPETITOR

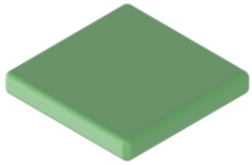
Competitors will be required to design, assemble, and program their robots during the training period before the competition day. Competitors will bring their robot fully assembled and functional.

Objects:

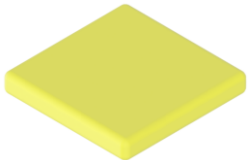
There are four objects with which the robots must interact.



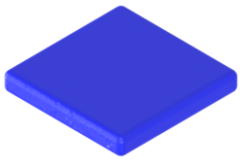
Red Flat Piece



Green Flat Piece



Yellow Flat Piece



Blue Flat Piece

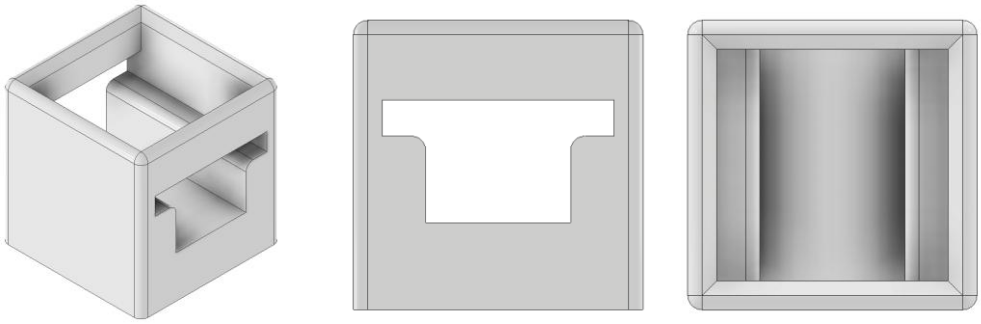
There is an additional object representing hazardous material. The object's color will be a surprise.

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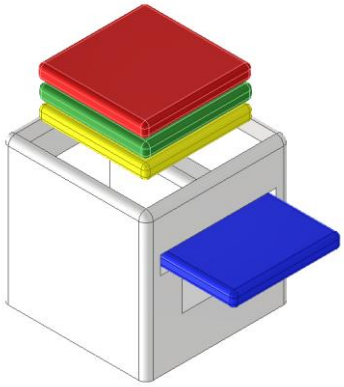
Surprise Flat Piece

The object will be provided to competitors during the morning briefing.

Storages:

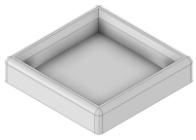
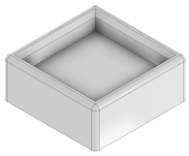
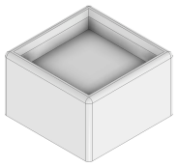
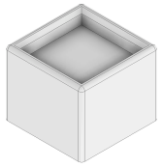


Flat objects are to be placed in storage. It is a competitor's responsibility to make sure the placement is correct before the start of the evaluation.



Delivery areas:

There are four bases with which the robots must interact in the delivery process.

			
25mm high	*50mm high*	*75mm high*	*100mm high*

Flat objects are to be placed on the bases in the different delivery areas according to the color.

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Robot work expectations are to take out the flat pieces from the storage base, identify the color of the flat piece by using a camera, and deliver the piece to complete the required task according to the order defined during the morning briefing.

Example of a robot task:

1. Start the robot by pressing the start button.
2. The robot drives to the storage area.
3. The robot drives to a position in front of a storage base.
4. The robot takes out the flat piece from the storage base.
5. Red flat piece taken.
6. The action for the red flat piece is to deliver it in one of the stands according to the order.
7. The robot transports it to the delivery areas.
8. The robot places the flat piece in the designated space on top of the stand.

EQUIPMENT, MACHINERY, INSTALLATIONS, AND MATERIALS REQUIRED

Teams can only use the Studica kit to develop their robots and complete the tasks.

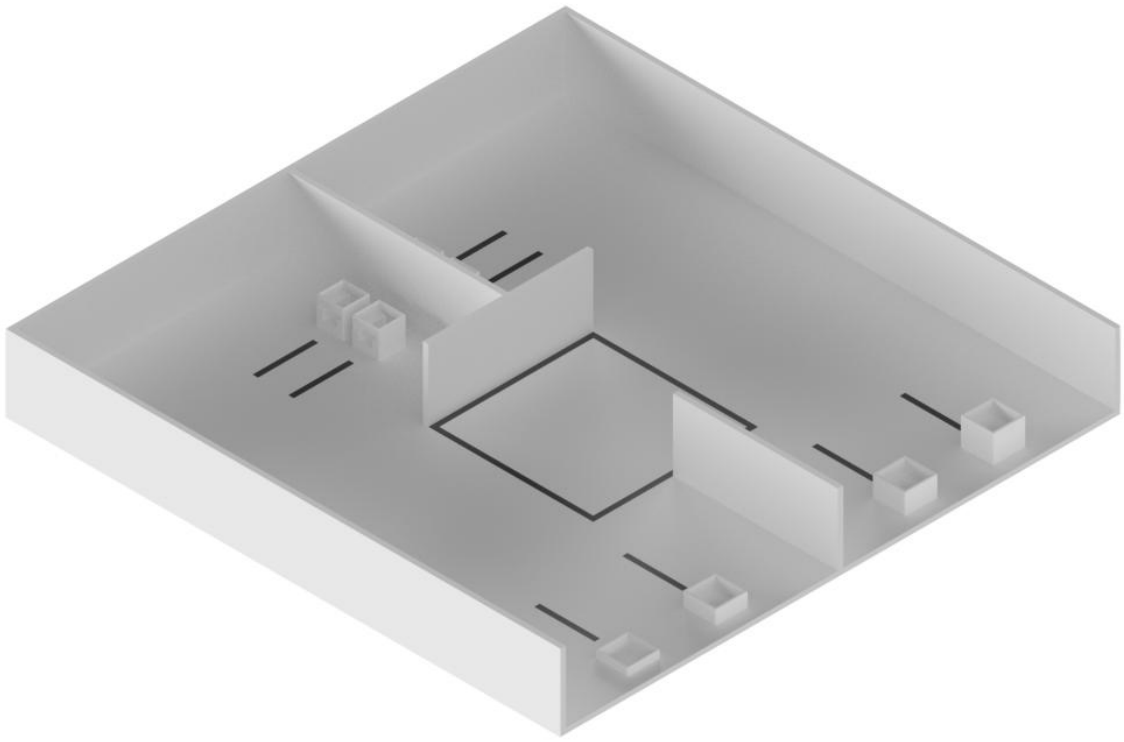
Competitors can incorporate “competitors designed / created components” into their robot design based on the following restrictions:

a. All sheet-based elements must be created using any polycarbonate material with a maximum overall sheet size of 1000mm by 1000mm by 4mm thick.
b. All 3D-printed elements must be created using ABS, PLA, PTU, Nylon, PETG, HIPS, ASA, or Carbon Filled Fiber with a maximum overall weight of 1.2 kg.

It is the responsibility of the competitors to provide replacement parts, both sheet-based parts and printed parts.

The arena required for this competition is shown below. The main features are an area of 2 x 2 meters with the inside walls, and the tapes on the floor to determine the areas.

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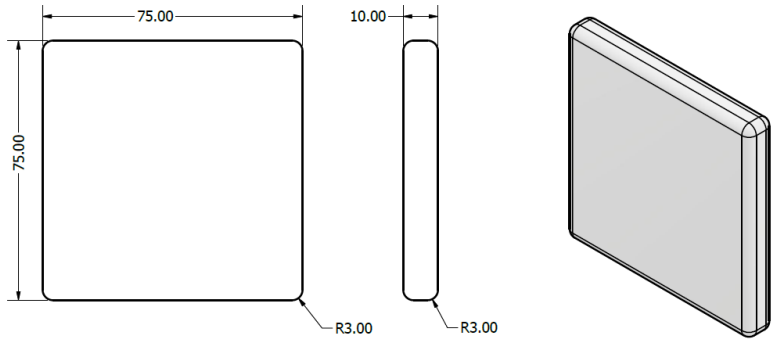
The campus host is responsible for providing the studica robot kit and necessary materials for the construction of the robot, all the necessary elements for the competition environment, including the robot arena, desks, chairs, computers, etc.

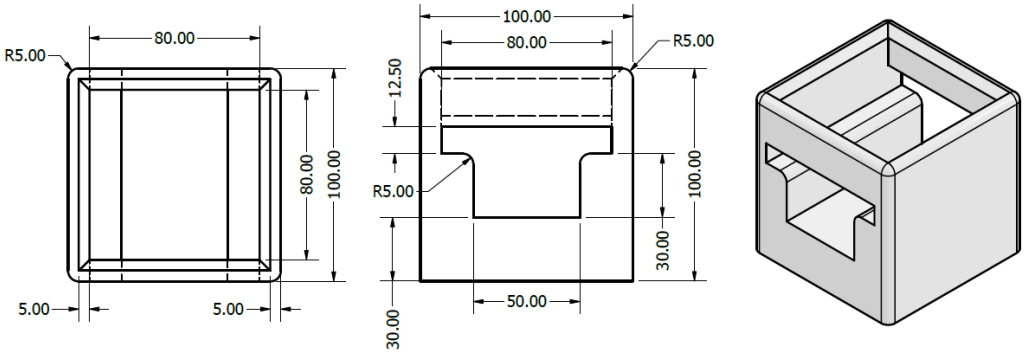
MARKING SCHEME

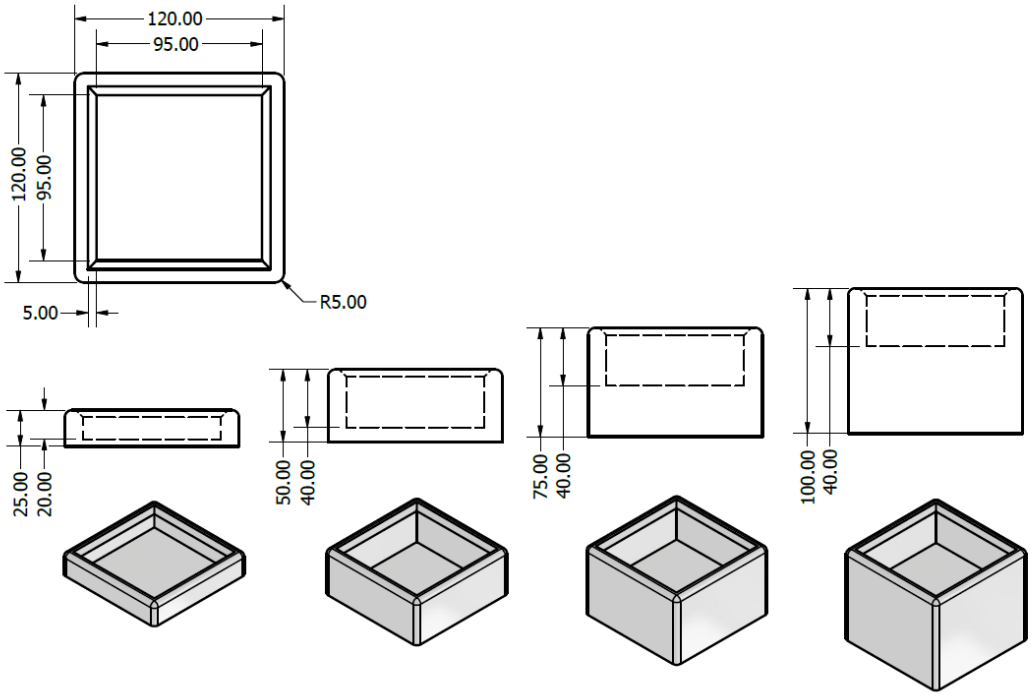
SECTION	CRITERION	JUDGEMENT MARKS	MEASUREMENT MARKS	TOTAL
A1	Work Organization & Behaviour	0.0	10.0	10.0
B1	Robot Design	0.0	10.0	10.0
C1	Simple Tasks	0.0	20.0	20.0
D1	Test Project Run Teleoperated	0.0	30.0	30.0
E1	Test Project Run Autonomous	0.0	30.0	30.0
TOTAL		0.0	100.0	100.0

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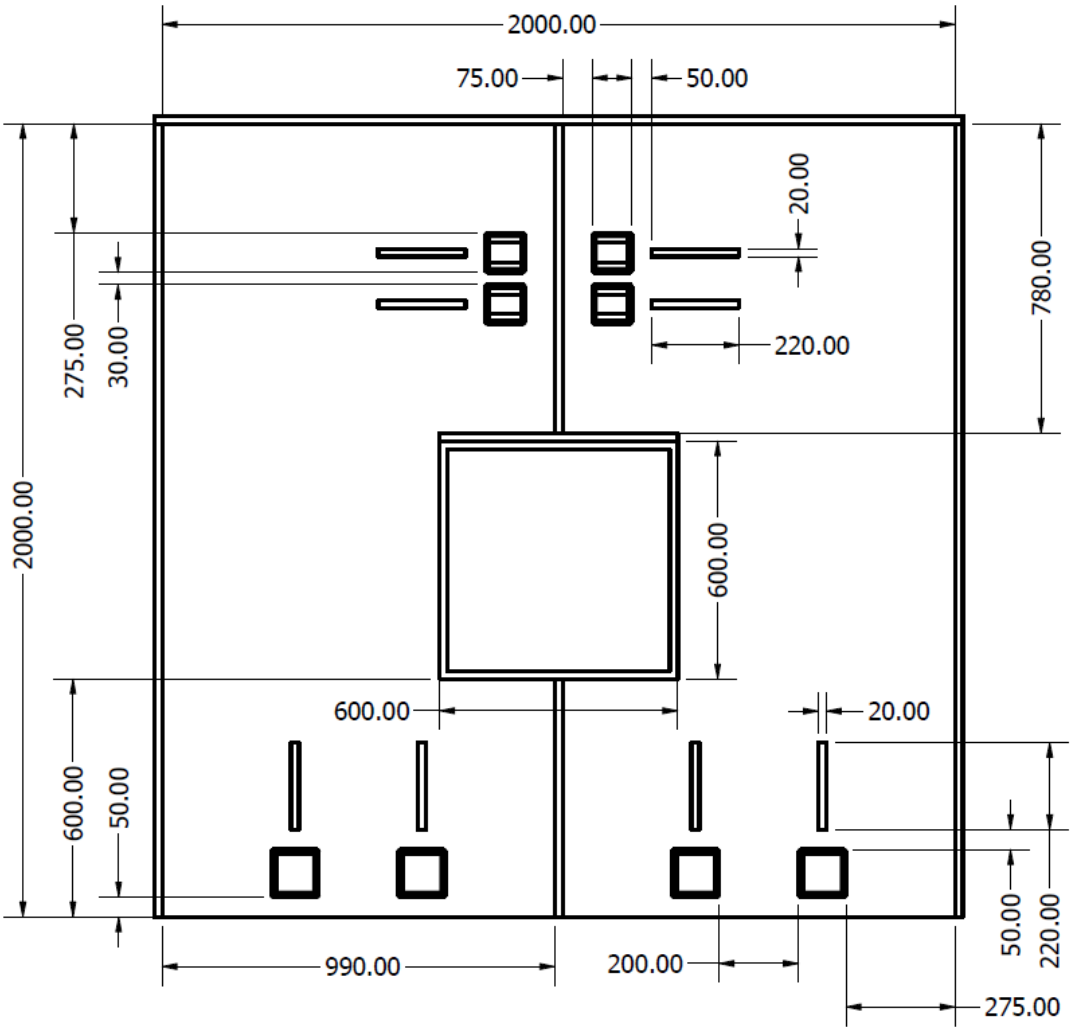
APPENDIX







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